

Integrable Systems — Exercise Sheet 15

The algebraic Bethe ansatz

Variant: Questions only

Conventions, as in Lecture 15: the Heisenberg chain of $N \geq 2$ sites, $L_{an}(\lambda) = \lambda + iP_{an}$, $T_a(\lambda) = L_{aN}(\lambda) \cdots L_{a1}(\lambda) = \begin{pmatrix} A & B \\ C & D \end{pmatrix}_a$, $t = A + D$; $|0\rangle$ is all spins up and $|n\rangle$ has one down spin, at site n ; $U(v_1 \otimes \cdots \otimes v_N) = v_N \otimes v_1 \otimes \cdots \otimes v_{N-1}$, $t(0) = i^N U$, and $H = \frac{J}{2} \sum_n (P_{n,n+1} - 1) = \frac{J}{2} (i \frac{d}{d\lambda} \ln t|_0 - N)$. Lecture 15 proves $A(\lambda)|0\rangle = (\lambda + i)^N |0\rangle$, $D(\lambda)|0\rangle = \lambda^N |0\rangle$, $C(\lambda)|0\rangle = 0$, $B(v)|0\rangle = i \sum_n v^{n-1} (v + i)^{N-n} |n\rangle$; with $f(u) = (u - i)/u$ and $g(u) = i/u$, the exchange relations $B(\lambda)B(\mu) = B(\mu)B(\lambda)$, $A(\lambda)B(\mu) = f(\lambda - \mu)B(\mu)A(\lambda) + g(\lambda - \mu)B(\lambda)A(\mu)$ and $D(\lambda)B(\mu) = f(\mu - \lambda)B(\mu)D(\lambda) - g(\lambda - \mu)B(\lambda)D(\mu)$ ($\lambda \neq \mu$); and, with $T_{11} = A$, $T_{12} = B$, $T_{21} = C$, $T_{22} = D$, the *RTT* relation entry by entry,

$$(\lambda - \mu) T_{jl}(\mu) T_{ik}(\lambda) - i T_{il}(\mu) T_{jk}(\lambda) = (\lambda - \mu) T_{ik}(\lambda) T_{jl}(\mu) - i T_{il}(\lambda) T_{jk}(\mu).$$

The momentum of an argument v is $e^{ik} = v/(v + i)$, and its rapidity $-v - \frac{i}{2}$.

Exercise 1 (one magnon). (i) For $v \neq 0, -i$ show that $B(v)|0\rangle$ is a plane wave, with amplitudes in the ratio $e^{ik} = v/(v + i)$ from site to site, and that $v = ie^{ik}/(1 - e^{ik})$ when $e^{ik} \neq 1$. (ii) Find $B(0)|0\rangle$ and $B(-i)|0\rangle$, and show that $B(v)|0\rangle / v^{N-1} \rightarrow iS^-|0\rangle$ as $v \rightarrow \infty$, where $S^- = \sum_n S_n^-$, so $S^-|0\rangle = \sum_n |n\rangle$. (iii) Using the exchange relations, show that for $\lambda \neq v$

$$t(\lambda)B(v)|0\rangle = \Lambda(\lambda)B(v)|0\rangle + \frac{i}{\lambda - v} [(v + i)^N - v^N] B(\lambda)|0\rangle,$$

$$\Lambda(\lambda) = (\lambda + i)^N f(\lambda - v) + \lambda^N f(v - \lambda),$$

and deduce that for $v \neq 0, -i$, $B(v)|0\rangle$ is an eigenvector of every $t(\lambda)$ if and only if $e^{ikN} = 1$. (iv) For such v , find the energy from Λ .

Exercise 2 (three sites by the algebraic Bethe ansatz). Take $N = 3$. (i) What are the eigenvalue of $t(\lambda)$ and the energy on $|0\rangle$? (ii) Find the regular solutions v of the Bethe equation for $M = 1$, their momenta k and rapidities, and show that $\Lambda(\lambda) = 2\lambda^3 + 3i\lambda^2 - ie^{-ik}$ on each. What are the energy and the eigenvalue of U ? (iii) Letting $v \rightarrow \infty$ in Exercise 1 (iii), show that $t(\lambda)S^-|0\rangle = [(\lambda + i)^N + \lambda^N]S^-|0\rangle$ for every N .

Exercise 3 (C past B, and a pairing). (i) From the *RTT* relation entry by entry, with $(i, j, k, l) = (2, 1, 1, 2)$ and $(2, 2, 1, 1)$, show that for $\lambda \neq \mu$

$$C(\lambda)B(\mu) - B(\mu)C(\lambda) = g(\lambda - \mu)[D(\lambda)A(\mu) - D(\mu)A(\lambda)],$$

and that $C(\lambda)C(\mu) = C(\mu)C(\lambda)$ for all λ, μ . (ii) Let $\langle 0|$ be the row vector of $|0\rangle$. Show that for $u \neq v$

$$\langle 0| C(u)B(v)|0\rangle = \frac{i}{u - v} [u^N (v + i)^N - v^N (u + i)^N].$$

(iii) Find the limit $u \rightarrow v$, and show that for a real rapidity it is $-\|B(v)|0\rangle\|^2$: the sign $(-1)^M$ of Lecture 15's reading on norms, at $M = 1$.

Exercise 4 (the singular pair). (i) Show that $\{v_1, v_2\} = \{0, -i\}$ satisfies Lecture 15's Bethe equations for $M = 2$, for every N . (ii) Show that $B(0) = i^N U S_N^-$, where S_N^- lowers the spin of site N . (iii) Deduce that $B(0)B(-i)|0\rangle = 0$ for every N , although neither factor alone annihilates $|0\rangle$. What does the algebraic Bethe ansatz give for this pair, and why is Lecture 15's energy formula stated for regular roots?

Exercise 5 (two creation operators). (i) Show that $f(a)g(b) + g(a)g(b-a) = g(b)f(a-b)$ for $a, b, a-b$ non-zero, and that the same identity holds for $F(u) = f(-u)$ and $G(u) = -g(u)$ in place of f and g . (ii) For pairwise distinct λ, v_1, v_2 , derive from the exchange relations

$$A(\lambda)B(v_1)B(v_2) = f(\lambda - v_1)f(\lambda - v_2)B(v_1)B(v_2)A(\lambda) + g(\lambda - v_1)f(v_1 - v_2)B(\lambda)B(v_2)A(v_1) \\ + g(\lambda - v_2)f(v_2 - v_1)B(\lambda)B(v_1)A(v_2),$$

and the same for D with $f(u) \rightarrow f(-u)$, $g(u) \rightarrow -g(u)$: the case $M = 2$ of Lecture 15's lemma. (iii) Acting on $|0\rangle$, find the unwanted coefficient of $B(\lambda)B(v_2)|0\rangle$ in $t(\lambda)B(v_1)B(v_2)|0\rangle$.

Exercise 6 (Baxter's T-Q relation). Let $Q(\lambda) = \prod_{j=1}^M (\lambda - v_j)$ with pairwise distinct v_j . (i) Show that Lecture 15's $\Lambda(\lambda) = (\lambda + i)^N \prod_j \frac{\lambda - v_j - i}{\lambda - v_j} + \lambda^N \prod_j \frac{\lambda - v_j + i}{\lambda - v_j}$ satisfies $\Lambda(\lambda)Q(\lambda) = (\lambda + i)^N Q(\lambda - i) + \lambda^N Q(\lambda + i)$. (ii) For $M = 1$, show that Λ is a polynomial if and only if $(v + i)^N = v^N$. (iii) For $N = 4$ the tutorial's ground state has $Q(\lambda) = \lambda^2 + i\lambda - \frac{1}{3}$. Without using its roots, find Λ by dividing the right side of the T-Q relation by Q , and read off K and E .

Exercise 7 ★. *The next charge, on magnons.* Lecture 15's reading on the tower of charges states that for $N \geq 4$, $\frac{d^2}{d\lambda^2} \ln t|_0 = N + \sum_n [P_{n,n+1}, P_{n+1,n+2}]$. (i) Let $v_1, \dots, v_M$ be regular roots of the Bethe equations with $\Psi \neq 0$, and $N \geq 3$. Show that on Ψ the operator $\frac{d^2}{d\lambda^2} \ln t|_0$ has the eigenvalue $N + \sum_j (2i \sin 2k_j - 4i \sin k_j)$: a sum of one function of each momentum. (ii) Compute $\sum_n [P_{n,n+1}, P_{n+1,n+2}]$ on $|x\rangle$, and hence on a magnon $\sum_x e^{ikx} |x\rangle$ with $e^{ikN} = 1$ and $e^{ik} \neq 1$, and check the reading's formula on it for $N \geq 4$. (iii) Check (i) on the four-site ground state of Exercise 6.